

MOHAMMED SEDEG

Senior AI Engineer · LLM Systems & Multi-Agent · Multimodal Deep Learning · MLOps

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PROFESSIONAL SUMMARY

AI Engineer with 5+ years building production LLM systems and edge computer vision, backed by 9 years of broader engineering across automation, embedded systems, and renewable energy. Across **6+ independent client engagements** since 2023, delivered **3× faster LLM inference** via GGUF/ExLlamaV2 quantization, **70% lower fine-tuning cost** via LoRA/QLoRA, and enterprise RAG platforms serving real-time workloads. Designs and ships **multi-agent LLM pipelines** end-to-end — architecture, evals, observability.

CORE COMPETENCIES

LLMs & Fine-Tuning

Hugging Face · LangChain · LoRA / QLoRA · RLHF / DPO · Multi-agent pipelines · LLM evals

Multimodal & Vision

LLaVA · Qwen-VL · YOLOv8 · ONNX / TFLite · OpenCV · Detectron2 · Jetson / Raspberry Pi

MLOps & Infrastructure

Docker · FastAPI · AWS SageMaker · MLflow · Prometheus · Grafana · Prefect · GitHub Actions

Languages

Python (advanced) · C/C++ · R · Bash · SQL

RAG & Vector Search

pgvector · Qdrant · Weaviate · FAISS · ChromaDB · MongoDB · Redis · Hybrid retrieval · Re-ranking

Training & Optimization

DeepSpeed · PyTorch DDP / FSDP · AMP · GGUF · ExLlamaV2 · TensorRT

ML & Research

PyTorch · TensorFlow · JAX · scikit-learn · XGBoost · LSTM · Transformer · CNN

Data & Storage

PostgreSQL · SQLAlchemy · PySpark · Alembic

PROFESSIONAL EXPERIENCE

Senior AI Engineer & LLM Consultant · Independent / Contract

2023 – Present

- **Enterprise RAG Architecture:** Shipped multi-provider LLM systems (OpenAI, Cohere, Hugging Face, Ollama) with modular retriever-generator pipelines over pgvector, Qdrant, and MongoDB via FastAPI — reducing query latency 40% on production workloads.
- **LLM Fine-Tuning at Scale:** Cut client fine-tuning costs 70% via LoRA and QLoRA on custom instruction datasets across 7B–13B parameter models; accelerated distributed training 20–30% using DeepSpeed, PyTorch DDP, and Automatic Mixed Precision.
- **Inference Optimization:** Deployed GGUF and ExLlamaV2 quantized models achieving 3× faster inference at under 50% memory footprint; built full observability stack (Prometheus + Grafana) and orchestrated pipelines with Prefect — cutting deployment cycle 35%.
- **Fintech KYC OCR Pipeline:** Built end-to-end document intelligence using Mistral OCR with structured extraction — cutting manual processing 60% for a financial-services client; improved reasoning 20% via Chain-of-Thought and Tree-of-Thought prompting.

AI Engineer — Computer Vision Specialist · VisionCore

2024 – 2025

- **Edge AI Deployment:** Shipped low-latency TFLite and ONNX models on <4 GB RAM hardware achieving 60–80 ms end-to-end inference for real-time industrial applications.
- **PPE Detection System:** Developed YOLOv8-based workplace safety detection reaching 85–90% mAP across 8 PPE categories; reduced overfitting 15% through advanced data augmentation (Mosaic, MixUp, Albumentations).
- **MLOps Automation:** Automated annotation and augmentation workflows via Roboflow and VLM-assisted labeling — cutting manual effort 25% and doubling dataset iteration speed.

- **Multimodal Solar Forecasting:** Designed an LSTM-Transformer-CNN fusion architecture over NOAA GOES-16/17 satellite imagery with cross-modal attention — outperforming published baselines on RMSE, MAE, and R^2 . *Manuscript currently under peer review.*
- **V2G Microgrid Optimization:** Built a Vehicle-to-Grid optimization pipeline using Improved Antlion Optimizer and PSO with Monte Carlo simulation — delivering full techno-economic benchmarking (COE, NPC, LPSP, REF) for renewable microgrid configurations.

- PLC/HMI/SCADA control systems (uptime +18%), solar deployments including a dual-axis tracker (+35% energy capture), and hybrid solar-EV R&D from concept to prototype.

LEADERSHIP & COMMUNITY

SAiR — Sudanese AI Research Initiative

 · Founder & Lead Instructor

- Founded and operate a free, open-source AI engineering program reaching **300+ learners across Sudan and Africa** with a 65% completion rate. Built and taught 6 structured modules spanning ML, deep learning, and MLOps; student teams have shipped production systems in healthcare diagnostics, crop disease detection, and Arabic NLP.

KEY PROJECTS

MyLLM

 — Transparent From-Scratch LLM Framework

- Installable pure-PyTorch framework covering the full LLM pipeline — tokenization → attention → training → RLHF → inference — with first-class support for GPT-2, LLaMA-2, and LLaMA-3. Includes one-line **LLM.from_pretrained()**, configurable SFT/DPO/PPO trainers, single-GPU/DDP/DeepSpeed/FSDP accelerators, and a CLI.
- Backed by **128 passing tests** (CPU-only, ~15s), **21 learning notebooks**, 5 Colab demos, and 34-page documentation — designed as a research-friendly alternative where every line is visible.

RagApp

 — End-to-End Retrieval-Augmented Generation System (v2)

- Production-ready async RAG platform on FastAPI with **dual vector DB support** (PostgreSQL + pgvector primary, Qdrant optional) switchable via a single env variable through a unified DB interface. Custom chunking logic (LangChain-optional), multi-provider embeddings/LLMs (OpenAI, Ollama, Cohere, Sentence Transformers), and Dockerized services.

Fero Cafe (Cafe.AI)

 — Production Multi-Agent Conversational Commerce System

- Full-stack 5-agent async pipeline (Guard → Classification → Details/Order/Recommendation) with RAG over local ChromaDB, SSE streaming, SQLite session persistence, and a live observability dashboard. Built on FastAPI + React Native (Expo), powered by Llama-3.3-70B via Groq with a provider-agnostic client. Shipped with 160 passing tests, LLM evals (80% pass-rate gating), and CI/CD via GitHub Actions.

V2G Microgrid Optimization & Solar Forecasting

 — Research Framework

- Python framework pairing V2G microgrid optimization (Improved Antlion Optimizer + PSO + Monte Carlo) with a multimodal LSTM-Transformer-CNN solar irradiance forecasting system.

EDUCATION

M.Sc. Mechatronics — Specialization: Computer Vision

2023

Karabuk University, Turkey

B.Sc. Electrical Engineering — Specialization: Control Systems

2016

Sudan University of Science and Technology

LANGUAGES Arabic — Native English — Fluent (C1) Turkish — Intermediate (B2)